NPEC × BUAS

# From root images to robot motion

Autonomous root tip targeting with computer vision,
PI control and reinforcement learning

***Arabidopsis thaliana*** • HADES / NPEC



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# The biological problem

Joint initiative of Wageningen University & Research and Utrecht University



CLIENT

**Netherlands Plant Eco-phenotyping Centre (NPEC)**

STUDY SPECIES

*Arabidopsis thaliana*

RESEARCH CONTEXT

**Plant–microbe interactions**

**The opportunity**

More consistent root inoculation could support larger experiments comparing responses across plant genotypes.



Sources: npec.nl/about-npec · dataset specimen imaging

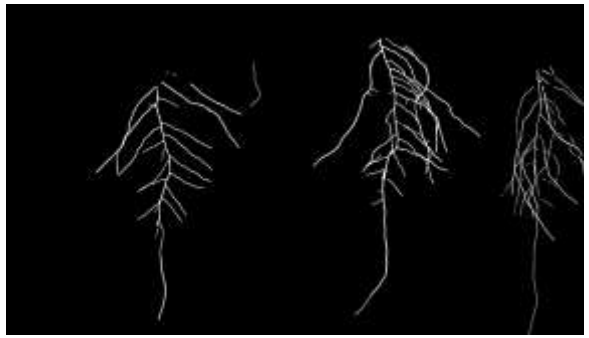
# One pipeline, three disciplines

Computer vision provides the target; robotics delivers the motion



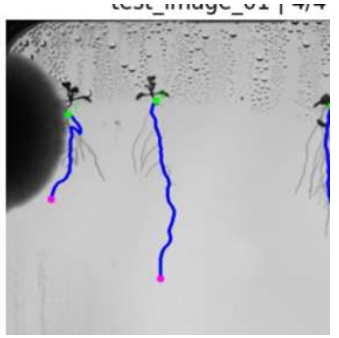
01 **HADES image**

Raw Petri dish



02 **Root segmentation**

512 px U-Net mask



03 **Root extraction**

Tip & length estimates

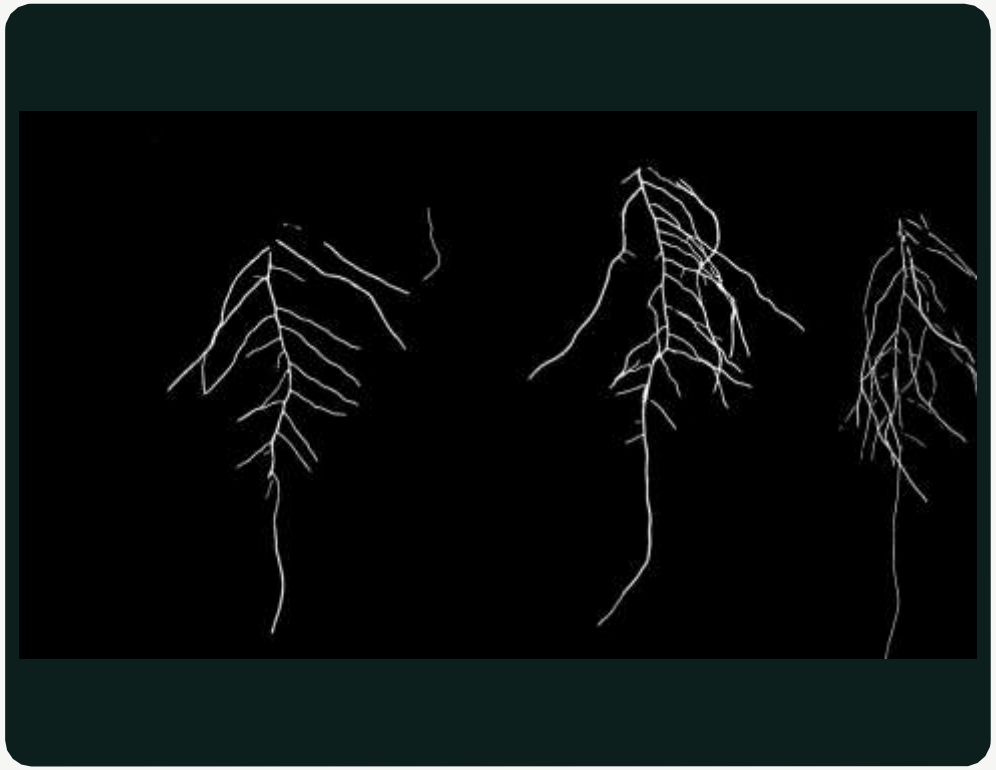
04 **Image coordinates**

05 **Assumed affine map**

06 **OT-2: PI / SAC**

# The 512 × 512 root segmentation model

Residual U-Net · group normalization · Dice + BCE + soft-clDice

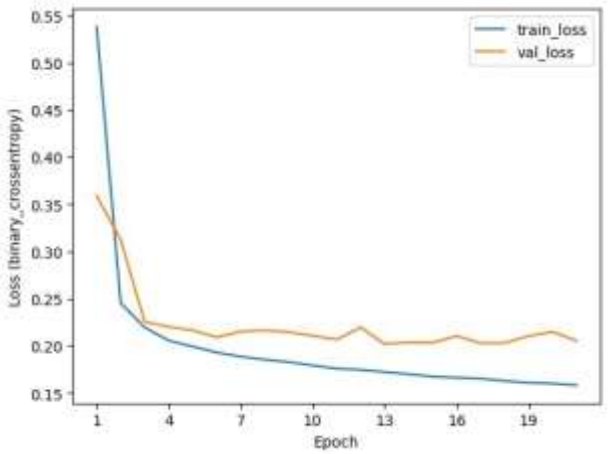


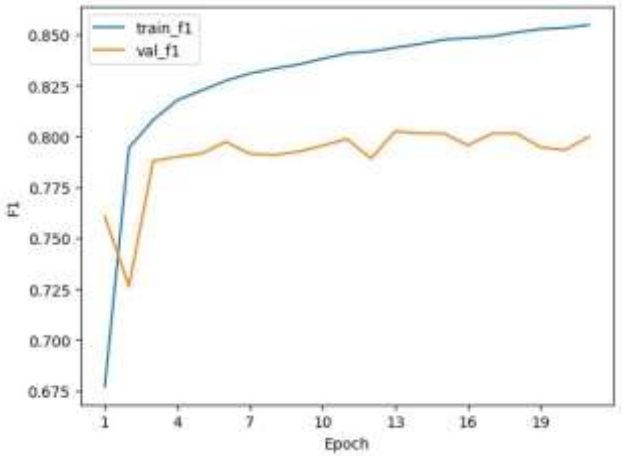
**0.8024**

**best validation F1**

**Train: 9,600 patches · Y2B_23 / _24**

Val: 612 patches · Y2B_25



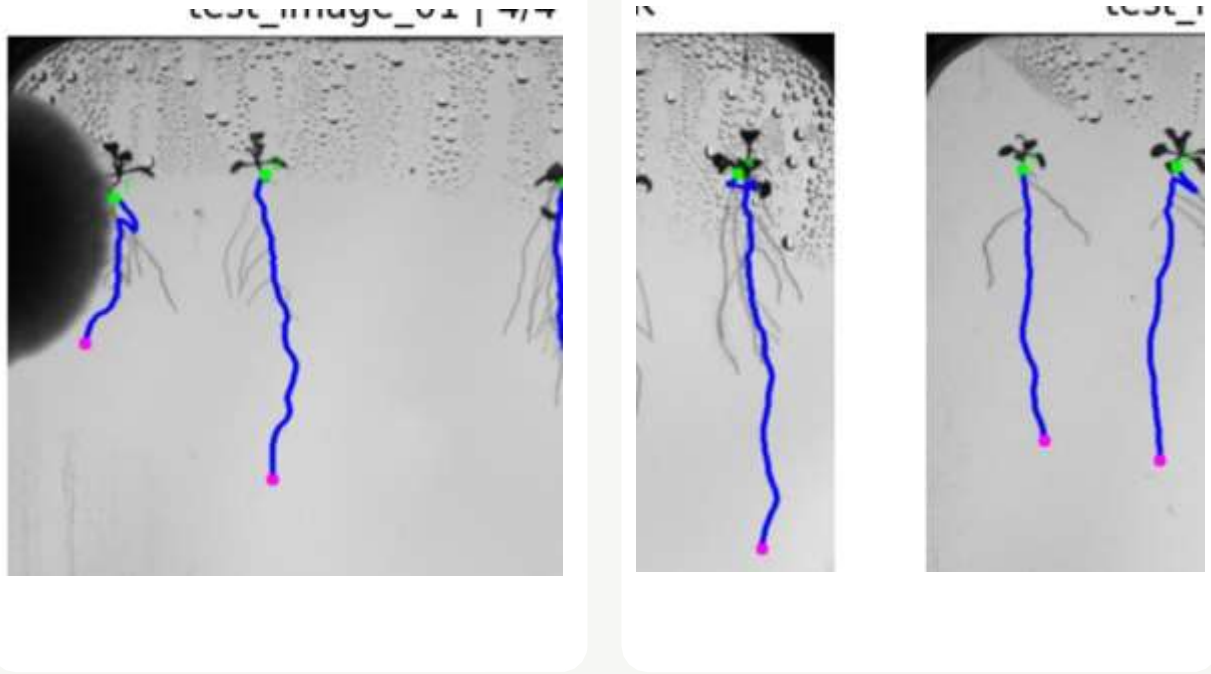

Faint or disconnected roots remain difficult, despite learned structure.



Source: Model training notebook + binary prediction image

# From a pixel mask to primary root measurements

Root-start refinement V5.1 → plant assignment → traced root length



**19/19**
images processed

**87**
root measurements

**11**
flagged for review

**76**
labelled OK

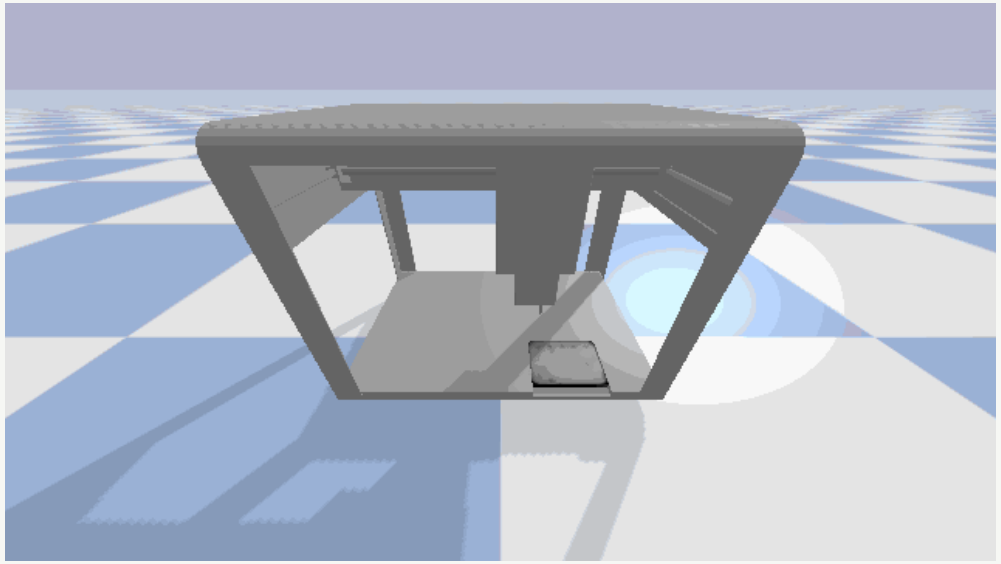
**16.159%**
FINAL PRIVATE KAGGLE sMAPE



Source: Task 8 notebook/README + Kaggle private screenshot

# Mapping root tips into the OT-2 workspace

CV gives pixel coordinates, image to plate transformation, Reachability + image-to-robot integration



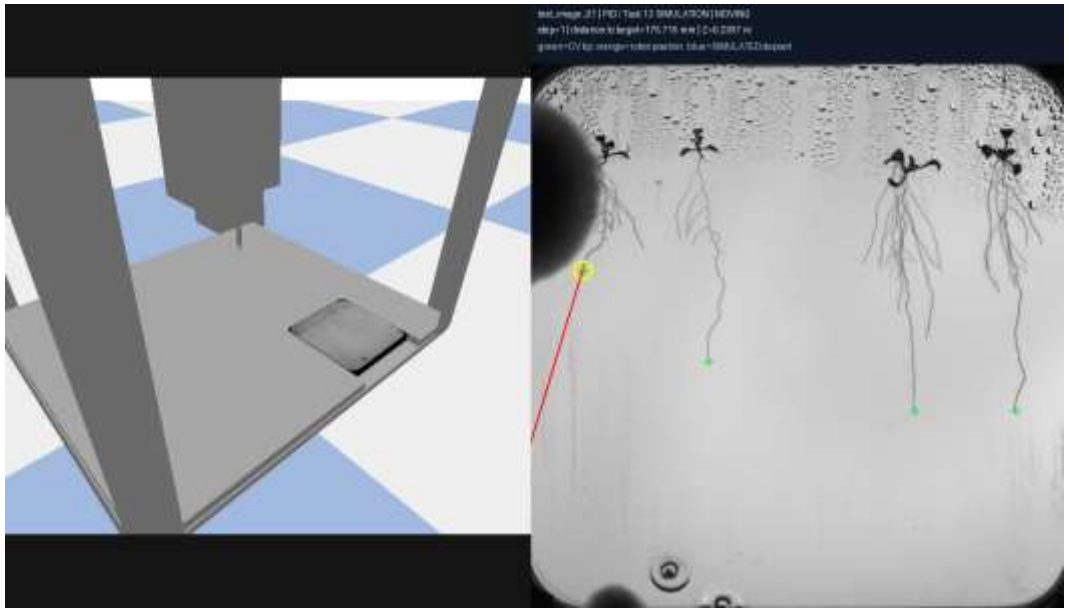
## WORKFLOW

1. Root-tip pixel (x, y)
2. Assumed image-to-plate transform
3. Mapped robot target (x, y, z)
4. **Approach · hover · retreat**

# Autonomous route demonstrations, pipeline in action

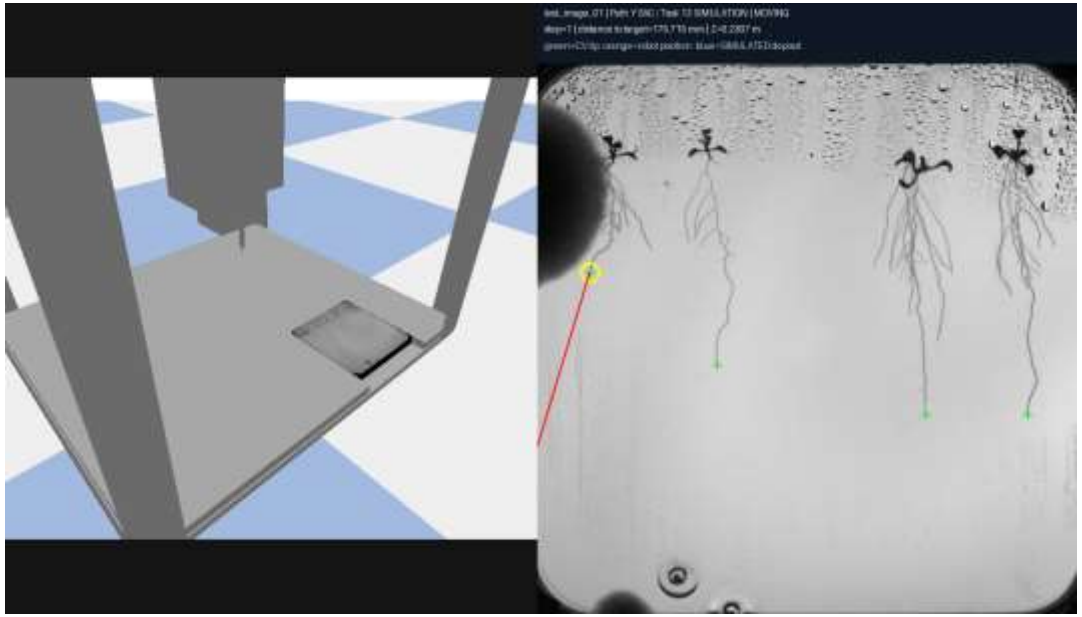
Three specimen images · 13 accepted computer-vision-derived targets

TUNED PI



**13/13 simulated route markers**

SAC POLICY



**9/13 simulated route markers**

Markers represent simulated success events, not liquid dispensing or biological inoculation.

# PI outperforms SAC on matched targets

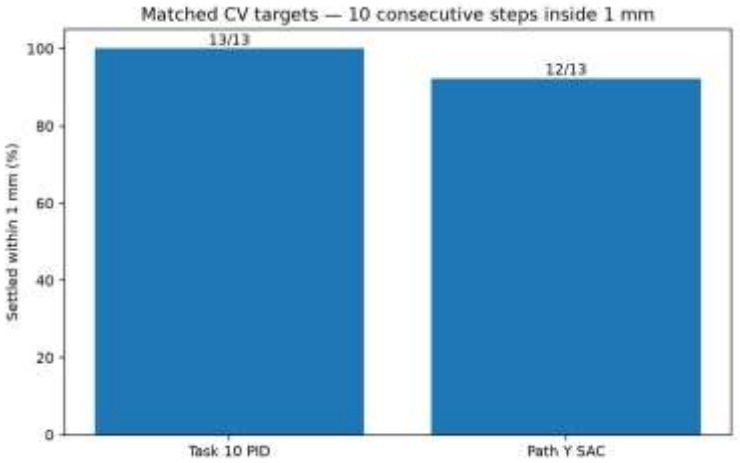
Identical 13 targets · reset pose · 600-step budget · 10-step dwell inside 1 mm

**13/13**
PI settled

**12/13**
SAC settled

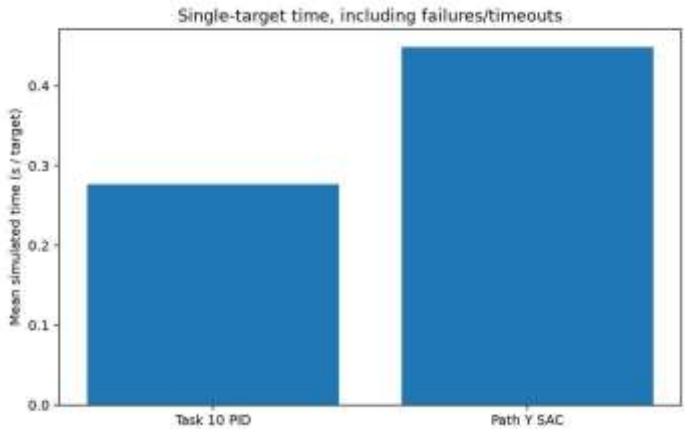
**0.179 mm**
PI mean final error

**0.561 mm**
SAC mean final error









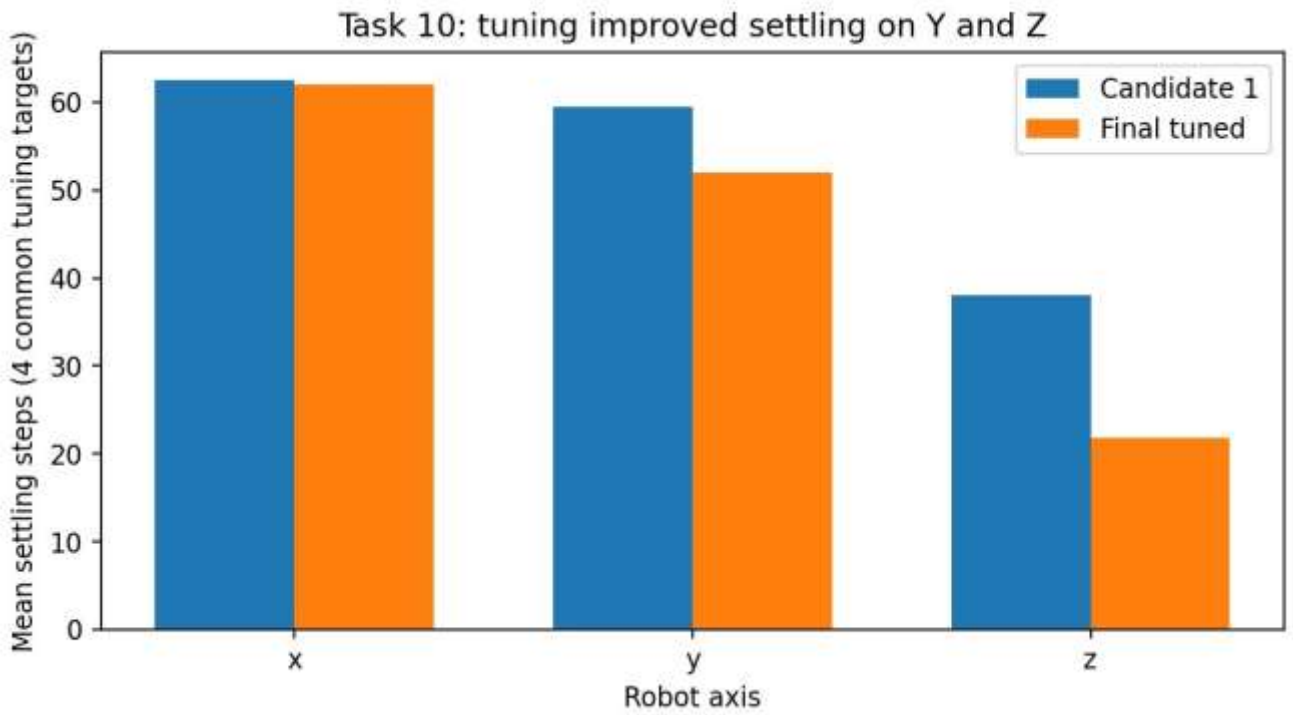

**Mean simulated time per target: PI 0.277 s • SAC 0.448 s | No unsafe terminations**



Source: matched_targets.csv + benchmark summary

# Tuning the PI positioning controller, XYZ axes

Used behavioural cloning, Error: different axes required different correction behaviour




**Y 59.5 → 52**

mean settling steps

**Z 38 → 21.75**

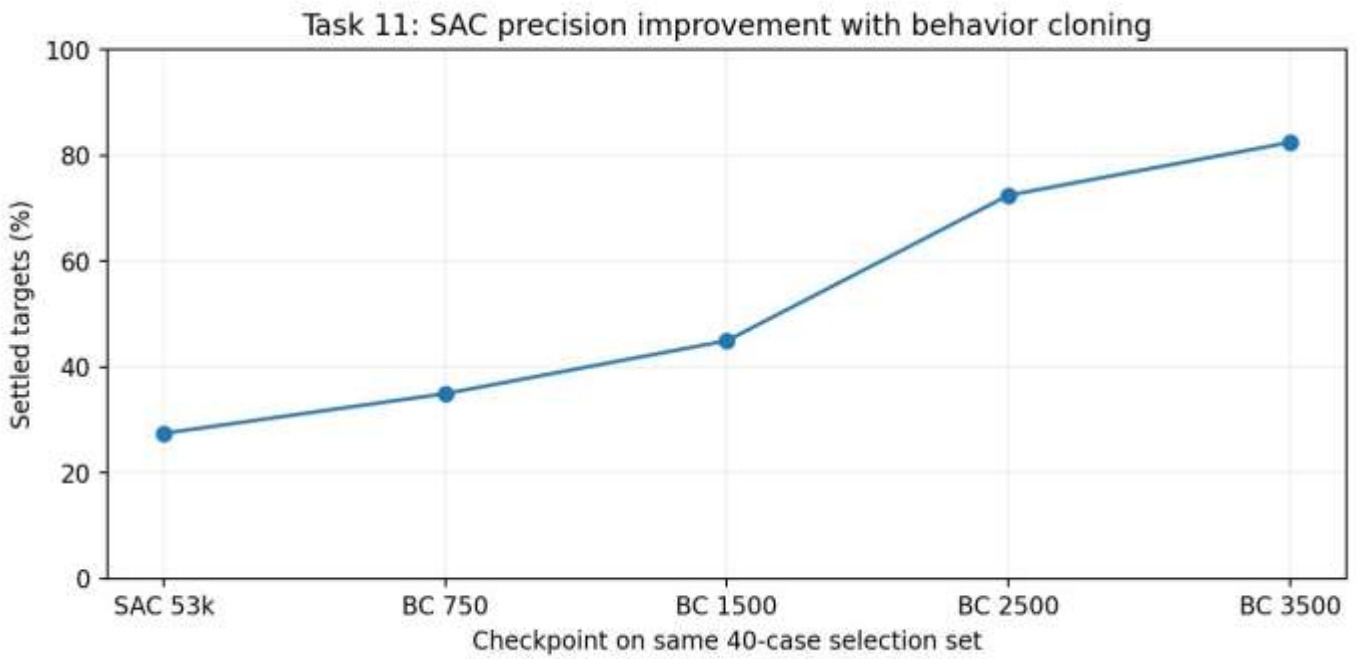
mean settling steps



Source: tuning_results.csv + gains_final.json

# SAC learned to approach, but not settle

Behavior cloning from successful controller trajectories improved precision




**11/40 → 33/40**

settled targets

**1.938 → 0.717 mm**

mean final error

**Holdout: 87/100 settled**

Additional guided RL did not beat BC.



Source: Path Y final_summary.json (same 40 selection cases)

# What the results do and do not show

Assumptions define the valid scope of the system

## ASSUMPTIONS

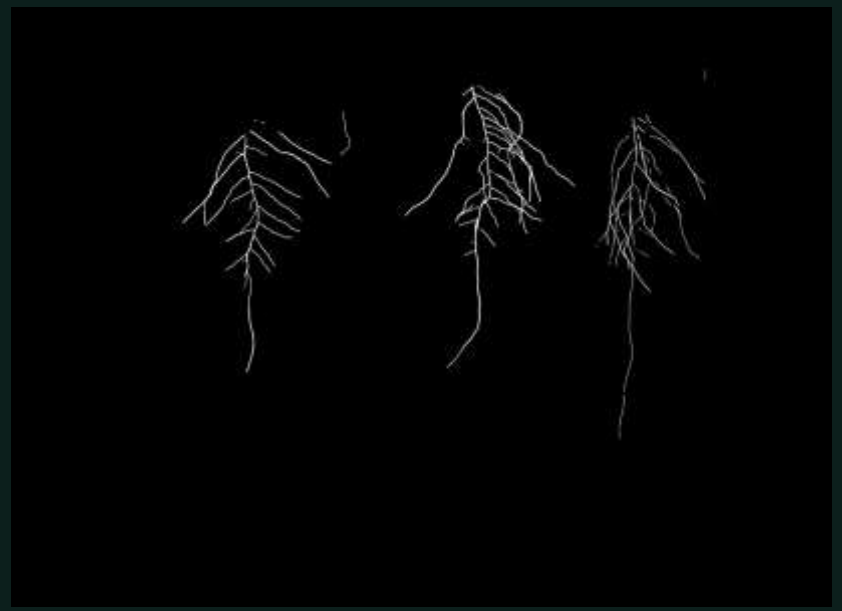
- Five seedlings in the seg image layout
- Similar image quality and lighting conditions
- Known plate position and dimensions in simulation

## LIMITATIONS

- Faint and overlapping root structures
- Only 13 selected integrated targets
- No measured physical registration
- No real droplet / infection validation

# The reliable controller today is PI.

The meaningful result is a working simulated pipeline from biological images to autonomous robot positioning.



| 13/13 | 0.179 mm | 0 |
|---|---|---|
| PI settled | PI mean error | unsafe episodes |

**NEXT: stronger root detection / measured calibration / real OT-2 deposition tests**

For NPEC: a foundation for scalable, consistent plant–microbe experiments.